Multivariate Calculus Lecture Notes
Section 12.3 – Dot Product

I. How do you multiply two vectors?
   a. One way is the *dot product* (or inner product).
   b. Definition of dot product
      i. Two dimensions: $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2$.
      ii. Three dimensions: $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$.
      iii. The definition extends to any n-dimensional vector.
   c. The dot product yields a scalar. The other way of multiplying vectors, the cross product, yields a vector.
   d. Examples
      i. $\langle 5, 4 \rangle \cdot \langle 2, 3 \rangle = 10 + 12 = 22$
      ii. $\langle -1, 0, 7 \rangle \cdot \langle 5, 2, 0 \rangle = -5$

II. Properties of dot product
   a. Commutative: $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
   b. Distributive: $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.
   c. $c(\mathbf{u} \cdot \mathbf{v}) = c\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot c\mathbf{v}$
   d. $\mathbf{0} \cdot \mathbf{v} = 0$
   e. $\mathbf{v} \cdot \mathbf{v} = ||\mathbf{v}||^2$
      i. In 3D, $\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + v_3^2 = ||\mathbf{v}||^2$.
      ii. This proof can be extended to any dimension.

III. Geometric interpretation of dot product
   a. The dot product is related to the angle between two vectors.
   b. Specifically, $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}|| \, ||\mathbf{v}|| \cos\theta$, where θ is the angle between the two vectors.
      i. See text for derivation.
      ii. Thus if $\mathbf{u}$ and $\mathbf{v}$ are unit vectors, then the dot product gives exactly the cosine of the angle between the two vectors.
   c. Example: $\mathbf{u} = \langle 2, 2, -1 \rangle$ and $\mathbf{v} = \langle 5, -3, 2 \rangle$
      i. $||\mathbf{u}|| = 3$
      ii. $||\mathbf{v}|| = \sqrt{38}$
      iii. $\mathbf{u} \cdot \mathbf{v} = 10 - 6 - 2 = 2$
      iv. Thus $\theta = \cos^{-1}(2/(3\sqrt{38}) \approx 1.46$ radians.
   d. What does it mean if the dot product is 0?
      i. If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\cos\theta = 0$. Thus $\theta = \pi/2$.
      ii. Therefore, if $\mathbf{u} \cdot \mathbf{v} = 0$ then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular).
      iii. The text actually makes this an "if and only if" statement by *defining* two vectors to be orthogonal when their dot product is 0.
   e. What does it mean if the dot product is negative?
      i. It means $\cos\theta < 0$.
      ii. Thus $\pi/2 < \theta < \pi$, and the angle between the vectors is obtuse.
   f. By convention, we take the angle between two vectors to be between 0 and π.

IV. Projections
   a. A projection of a vector onto another is essentially the shadow the first vector makes on the second.
   b. Notation: $\mathbf{u}_{||\mathbf{v}}$, the projection of $\mathbf{u}$ onto $\mathbf{v}$.

c. Example
   i. Project $\langle 5, 4\rangle$ onto x-axis. You get $\langle 5, 0\rangle$.
   ii. Project $\langle 5, 4\rangle$ onto y-axis. You get $\langle 0, 4\rangle$.

d. You can do this with any pair of vectors.
   i. For example, one can project $\langle 5, 4\rangle$ onto $\langle 1, 1\rangle$.
   ii. Show this graphically.
   iii. How do you figure this out?

e. Derivation of projection formula
   i. We need to express $\mathbf{u}_{||\mathbf{v}}$ in terms of its magnitude times a unit vector in the appropriate direction.
   ii. Let $\mathbf{e_v}$ denote the unit vector in the direction of $\mathbf{v}$.
   iii. Sign issues
      1. If θ is acute
         • Geometrically:

         • We have

$$\mathbf{u}_{||\mathbf{v}} = ||\mathbf{u}|| \cos\theta \mathbf{e_v} = ||\mathbf{u}|| \cos\theta \frac{\mathbf{v}}{||\mathbf{v}||}.$$

      2. If θ is obtuse
         • Geometrically:

         • We have

$$\mathbf{u}_{||\mathbf{v}} = -||\mathbf{u}|| \cos\theta \frac{-\mathbf{v}}{||\mathbf{v}||} = ||\mathbf{u}|| \cos\theta \frac{\mathbf{v}}{||\mathbf{v}||}.$$

      3. Either way we have the same expression for $\mathbf{u}_{||\mathbf{v}}$.
   iv. We also know that $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||\,||\mathbf{v}|| \cos\theta$.
   v. Substitution yields

$$\mathbf{u}_{||\mathbf{v}} = ||\mathbf{u}|| \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||\,||\mathbf{v}||} \frac{\mathbf{v}}{||\mathbf{v}||} = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\mathbf{v}, \text{ or}$$
$$\mathbf{u}_{||\mathbf{v}} = (\mathbf{u} \cdot \mathbf{e_v})\mathbf{e_v}.$$

f. Verify formula for $\mathbf{u} = \langle 5, 4\rangle$:
   i. $\text{Proj}_{\mathbf{i}}\mathbf{u} = 5\mathbf{i}$.
   ii. $\text{Proj}_{\mathbf{j}}\mathbf{u} = 4\mathbf{j}$.

g. Vector components
   i. The vector $\mathbf{u} = \langle 5, 4\rangle$ can be expressed as $\mathbf{u}_{||\mathbf{v}} + \mathbf{u}_{\perp\mathbf{v}}$, where $\mathbf{u}_{||\mathbf{v}}$ is as defined before, and $\mathbf{u}_{\perp\mathbf{v}}$ is the vector component of $\langle 5, 4\rangle$ orthogonal to $\langle 1, 1\rangle$.
   ii. Graphically:

h. Example: $\mathbf{u} = \langle 5, 4\rangle$ and $\mathbf{v} = \langle 1, 1\rangle$.
   i. $\mathbf{u}_{||\mathbf{v}} = \frac{9}{2}\langle 1, 1\rangle = \langle \frac{9}{2}, \frac{9}{2}\rangle$. (Note that this must be a multiple of $\langle 1, 1\rangle$!)
   ii. Then $\mathbf{u}_{\perp\mathbf{v}} = \langle 5, 4\rangle - \langle \frac{9}{2}, \frac{9}{2}\rangle = \langle \frac{1}{2}, -\frac{1}{2}\rangle$.
   iii. Check: $\mathbf{u}_{||\mathbf{v}} \cdot \mathbf{u}_{\perp\mathbf{v}} = 0$.

V. Final example: Find the angle θ between the diagonal of a cube and one of its edges.
   a. Graphically:

   b. It doesn't matter which diagonal and which edge we use; the angle stays the same.
   c. A vector representing the diagonal of the cube is $\langle 1, 1, 1\rangle$.
   d. A vector representing the edge of the cube in the x direction is $\langle 1, 0, 0\rangle$.
   e. So we have
$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||\,||\mathbf{v}||} = \frac{1}{\sqrt{3}}.$$
   f. Thus $\theta \approx 0.955$ radians.

Homework: Section 12.3 #2, 10, 24, 25, 34, 35, 49, 56, 64, 68, 76.